

Reachy Mini Robot & Baymax Setup Guide

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Phase 1: Find the Robot's (Pi Zero 2W) IP Address

Make sure the robot (Pi Zero 2W) and your Jetson Orin Nano are connected to the same local Wi-Fi network.

1. Open a terminal on your Jetson Orin Nano and run:
`ping -c 3 raspberrypi.local`
(If you custom-named the robot, try: `ping -c 3 reachy.local`)
2. Copy the IP address printed in the ping response (e.g., 192.168.1.45).
3. If the ping fails, scan the local network to find the IP:
`arp -a`

Phase 2: Start the Robot Server (Raspberry Pi Zero 2W)

1. SSH into the Pi Zero 2W using the IP address:
`ssh pi-zero@192.168.1.22`
2. Navigate to the Pi robot server directory:
`cd /reachy_mini/Reachy_mini_custom/reachy_mini_v6/pi_robot`
3. Start the socket and camera streams:
`python3 robot_server.py`

Keep this terminal window running. This initializes the Motor Server (port 5001), Mic Server (port 5002), Speaker Server (port 5003), and background Camera Stream (port 5000).

Phase 3: Set up the Voice Assistant (Jetson Orin Nano)

Open a new terminal window on your Jetson Orin Nano.

1. Clone the project and configure the environment:
`git clone https://github.com/vivinarya/small-voice.git`
`cd small-voice`
`python3 -m venv jarvis-env`
`source jarvis-env/bin/activate`
2. Install python packages:
`pip install -r requirements-orin.txt`
3. Download local Piper TTS assets:
`mkdir -p assets/piper/ assets/piper_voices/`
`wget https://github.com/rhasspy/piper/releases/download/v1.2.0/piper_arm64.tar.gz`

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tar -xf piper_arm64.tar.gz
cp piper/piper assets/piper/
chmod +x assets/piper/piper
wget -O assets/piper_voices/en_US-lessac-medium.onnx https://huggingface.co/rhasspy/piper-voices/resolve/main/en/en_US/lessac/medium/en_US-lessac-medium.onnx
wget -O assets/piper_voices/en_US-lessac-medium.onnx.json https://huggingface.co/rhasspy/piper-voices/resolve/main/en/en_US/lessac/medium/en_US-lessac-medium.onnx.json
4. Setup Ollama (Local LLM Engine):
curl -fsSL https://ollama.com/install.sh | sh
ollama pull qwen2.5:3b
5. Update config.yaml with the Robot IP:
cp config-orin.yaml config.yaml
Open config.yaml in a text editor (e.g. nano config.yaml) and set:
robot:
  enabled: true
  ip: "<YOUR_ROBOT_IP>"
```

Phase 4: Run the Assistant

1. Start the conversational assistant on the Jetson Orin Nano:
python3 src/main.py
2. Trigger the dialogue by saying `***Hey Baymax***` (or `***Baymax***`).
3. Ask your question. Speech synthesized on the Jetson will play directly from the robot speakers (port 5003), and the robot will perform talking sway sways (port 5001) in real-time.